

WELL COMPLETION REPORT (SYNTHETIC)

OIL-HPJ-009 - Hapjan-009

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000103
Field / District	Hapjan / Tinsukia
Block	OALP-III-07
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2017-08-06 / 2017-10-07
Measured / TVD	2406 m / 2364 m
Trajectory	Directional; inc 16.6 deg; az 42.1 deg
Formation Interval	1133-1990 m
Lithology	Argillaceous sandstone and shale
Pore Pressure Reference	2251 psi
Fracture Gradient	14.10 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Low (38.0/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	601	Surface
9 5/8 in casing	1395	Intermediate
7 in liner/casing	2117	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2017-09-27 2044 m	NPT	Critical	No depth progress occurred during the recorded interval. Mitigation: Use preventive maintenance triggers from repeated downtime patterns. Lesson: Separate geological risk from equipment-driven NPT in analytics.
2017-08-22 755 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Offset loss history should inform cementing hydraulics.
2017-08-29 1011 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2017-09-17 1713 m	NPT	High	ROP dropped to zero during equipment unavailability. Mitigation: Use preventive maintenance triggers from repeated downtime patterns. Lesson: Separate geological risk from equipment-driven NPT in analytics.
2017-08-20 694 m	Pressure Spike	Medium	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2017-08-27 933 m	Torque Spike	Medium	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2017-09-03 1204 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 2406 m MD in the Girujan interval. The NWIS knowledge record links 10 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.